

Lecture 4

Time and Change

There is a massive, quiet, intimidating man sitting alone at the end of the bar. His T-shirt says "−1."

Art: *Who is that "Minus One" guy over in the corner? The bouncer?*

Lenny: *He's way more than a bouncer. He's*

THE LAW.

Without him, this whole place would fall apart.

4.1 A CLASSICAL REMINDER

In *Volume I*, it took a little more than a page to explain what a state is in classical mechanics. The quantum version has taken three lectures, three mathematical interludes, and according to my rough count, about 17,000 words to get to the same place. But I think the worst is over. We now know what a state is. However, just as in classical physics, knowing the states of a system is only half the story. The other half involves a rule about how states change with time. That's our next job.

Let me just give you a quick reminder about the nature of change in classical physics. In classical physics, the space of states is a mathematical